

Detailed Theory

1. Configure and Verify PPP and MLPPP on WAN Interfaces Using Local Authentication

PPP (Point-to-Point Protocol):

PPP is a data link layer protocol commonly used on serial WAN links. It provides encapsulation of network layer protocols, link control, authentication, and error detection.

Key Features:

- Supports multiple Layer 3 protocols (IP, IPX, etc.)
- Includes Link Control Protocol (LCP) for link establishment, configuration, and testing.
- Authentication via PAP (Password Authentication Protocol) or CHAP (Challenge Handshake Authentication Protocol).

Configuration Steps for PPP with Local Authentication:

- Enable PPP encapsulation on the serial interface.
- Configure authentication method (usually CHAP for security).
- Create local username and password for authentication.
- Enable the authentication on the interface.

MLPPP (Multilink PPP):

MLPPP allows the bundling of multiple physical WAN links into a single logical link, increasing bandwidth and redundancy.

Configuration:

- Define a multilink interface.
- Bind multiple physical interfaces to the multilink.

- Enable PPP encapsulation and authentication on each physical interface.
- Configure the multilink interface with IP addressing.
- Use local authentication as with PPP.

Verification Commands:

- show interfaces serial X/X — shows interface status.
- show ppp multilink — displays MLPPP stats.
- show authentication sessions — shows authentication status.
- debug ppp negotiation — troubleshooting.

2. Configure, Verify, and Troubleshoot PPPoE Client-Side Interfaces Using Local Authentication

PPPoE (PPP over Ethernet):

PPPoE encapsulates PPP frames inside Ethernet frames. Commonly used in DSL broadband connections where a subscriber connects via Ethernet but authenticates with PPP.

Key Points:

- Client initiates PPP session over Ethernet.
- Supports PAP and CHAP for authentication.
- Requires virtual-template interface configuration on the router.

Configuration Steps:

- Create a virtual-template interface with PPP encapsulation and authentication.
- Configure local user credentials.
- Configure physical interface to use PPPoE client.
- Bind the dialer interface for PPPoE.

Verification:

- show pppoe session — checks PPPoE session status.
- show interfaces dialer — checks dialer interface.
- debug ppp negotiation — for authentication troubleshooting.

3. Configure, Verify, and Troubleshoot GRE Tunnel Connectivity

GRE (Generic Routing Encapsulation):

GRE tunnels encapsulate packets from any Layer 3 protocol inside IP tunnels, allowing for point-to-point connections over an IP network.

Key Features:

- Supports encapsulation of multiple protocols.
- Creates virtual point-to-point links.
- Can be used to connect remote networks over the internet.

Configuration Steps:

- Define tunnel interfaces on both routers.

- Specify source and destination IP addresses for tunnel endpoints.
- Assign IP addresses to tunnel interfaces.
- Configure routing to use the tunnel interface.
- Verify connectivity with ping and traceroute.

Verification Commands:

- show interfaces tunnel – status of tunnel.
- show ip route – check tunnel routes.
- show ip interface brief – verify IP assignment.

- debug tunnel — troubleshooting.

4. Configure and Verify Single-Homed Branch Connectivity Using eBGP IPv4 (Peering and Route Advertisement Using Network Command Only)

eBGP (External Border Gateway Protocol):

eBGP is used to exchange routing information between different autonomous systems (AS).

Key Points:

- Establish peering between ASes.
- Advertise networks using the network command.
- Requires neighbor configuration with remote AS.

Configuration Steps:

- Configure BGP with local AS.
- Define eBGP neighbor IP and remote AS.
- Advertise connected networks using network command.
- Verify peering status and routes.

Verification Commands:

- show ip bgp summary – BGP neighbor status.
- show ip bgp – BGP routing table.

- show ip route bgp — routes learned via BGP.
- debug ip bgp — for troubleshooting.